

CSE 102 (Fall 2021) – Homework 1

Due: Oct 5 at 10pm PT on CrowdGrader.

1. Let $f(n)$ and $g(n)$ asymptotically positive functions. Prove that $f(n) + g(n) = \Theta(\max(f(n), g(n)))$.

Solution. *Note:*

$$f(n) \leq f(n) + g(n) \quad (1)$$

$$g(n) \leq f(n) + g(n) \quad (2)$$

Therefore:

$$\max(f(n), g(n)) \leq f(n) + g(n) \quad (3)$$

$$\max(f(n), g(n)) = O(f(n) + g(n)) \quad (4)$$

since:

$$O(g(n)) = 0 \leq f(n) \leq c \cdot g(n) \quad (5)$$

This also implies that:

$$f(n) + g(n) \in \Omega(\max(f(n), g(n))) \quad (6)$$

using $f(n) = O(g(n))$ iff $g(n) = \Omega(f(n))$ and vice-versa. (Proven in class)

Then:

$$f(n) + g(n) \leq 2 \cdot \max(f(n), g(n)) \quad (7)$$

$$f(n) + g(n) = O(\max(f(n), g(n))) \quad (8)$$

since:

$$O(g(n)) = 0 \leq f(n) \leq c \cdot g(n) \quad (9)$$

From this we can conclude that:

$$f(n) + g(n) \in \Theta(\max(f(n), g(n))) \quad (10)$$

because:

$$f(n) + g(n) \in \Omega(\max(f(n), g(n))) \quad (11)$$

$$f(n) + g(n) \in O(\max(f(n), g(n))) \quad (12)$$

$$\Theta(g(n)) = O(g(n)) \cap \Omega(g(n)) \quad (13)$$

2. Prove or disprove: If $f(n) = \Theta(g(n))$, then $f(n)^2 = \Theta(g(n)^2)$.

Solution. *If $f(n) = \Theta(g(n))$ is true then $f(n)$ and $g(n)$ have the same order of growth. This in turn tells us that $f(n)^2$ has the same order of growth as $g(n)^2$. Therefore the statement should hold for any*

$f(n)$ and $g(n)$ as long as both are of the same order of growth. Lets take $f(n) = 2n$ and $g(n) = n$ as an example.

$$f(n)^2 = \Theta(g(n)^2) \quad (14)$$

$$2n^2 = \Theta(n^2) \quad (15)$$

$$4n^2 \in \Theta(n^2) \quad (16)$$

or lets take $f(n) = 5 \cdot \log_2(n)$ and $g(n) = \log_2(n)$ as an example.

$$f(n)^2 = \Theta(g(n)^2) \quad (17)$$

$$5 \cdot \log_2(n)^2 = \Theta(\log_2(n)^2) \quad (18)$$

$$25 \cdot n \in \Theta(n) \quad (19)$$

3. Prove or disprove: If $f(n) = \Theta(g(n))$, then $2^{f(n)} = \Theta(2^{g(n)})$.

Solution. Lets take $f(n) = 2n$ and $g(n) = n$. $f(n) = \Theta(g(n))$ holds true however if we continue:

$$2^{f(n)} = \Theta(2^{g(n)}) \quad (20)$$

$$2^{2n} = \Theta(2^n) \quad (21)$$

$$4^n \neq \Theta(2^n) \quad (22)$$

We can see that for this case $2^{f(n)} = \Theta(2^{g(n)})$ is not true. Therefore the statement has been disproven.

4. Let $f(n)$ and $g(n)$ be asymptotically positive functions, and assume that $\lim_{n \rightarrow \infty} g(n) = \infty$. Prove that if $f(n) = \Theta(g(n))$, then $\ln(f(n)) = \Theta(\ln(g(n)))$.

Solution.

$$f(n) = \Theta(g(n)) \text{ if } 0 < \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} < \infty \quad (23)$$

$$\text{If } \lim_{n \rightarrow \infty} g(n) = \infty \text{ then } \lim_{n \rightarrow \infty} f(n) = \infty \quad (24)$$

$$\text{Then: } \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{\infty}{\infty} = 1 \quad (25)$$

$$(26)$$

This means that $f(n)$ and $g(n)$ have the same order of growth. Then let $f(n) = 2n$ and $g(n) = n$:

$$\ln(f(n)) = \Theta(\ln(g(n))) \quad (27)$$

$$\ln(2n) = \Theta(\ln(n)) \quad (28)$$

$$2 \cdot \ln(n) \in \Theta(\ln(n)) \quad (29)$$

5. Show that if $f(n)\ln(f(n)) = \Theta(n)$, then $f(n) = \Theta(\frac{n}{\ln(n)})$. Hint: use the result of the preceding problem.

Solution.

$$f(n)\ln(f(n)) = \Theta(n) \quad (30)$$

$$f(n) = e^{\frac{\Theta(n)}{f(n)}} \quad (31)$$

$$f(n) = e^{\frac{\Theta(n)}{\Theta(\frac{n}{\ln(n)})}} \quad (32)$$

$$f(n) = e^{\Theta(\frac{n}{\ln(n)})} \quad (33)$$

$$f(n) = e^{\Theta(\ln(n))} \quad (34)$$

$$\ln(f(n)) = \Theta(\ln(n)) \quad (35)$$

Since $\ln(f(n)) = \Theta(\ln(n))$ is true then the above statement is also true.

6. Consider the statement: $f(cn) = \Theta(f(n))$.

A Determine a function $f(n)$ and a constant $c > 0$ for which the statement is false.

Solution. Let $f(n) = 2^n$ and $c = 2$

$$f(n) = 2^n \quad (36)$$

$$f(cn) = f(2n) = 2^{2n} = 4^n \quad (37)$$

$$4^n \neq \Theta(2^n) \quad (38)$$

$$(39)$$

The above statement does not hold if $f(n) = 2^n$ and $c = 2$.

B Determine a function $f(n)$ for which the statement is true for all $c > 0$.

Solution. If $f(cn) = \Theta(f(n))$ that means that $f(cn)$ is of the same order of growth as $f(n)$.

Due to this as long as $f(n) = n^x$ where $x \geq 1$ the statement should hold.

Let $f(n) = n^2$ and $c = 2$

$$f(n) = n^2 f(cn) = f(2n) = 2n^2 = 4n^2 \in \Theta(n^2) \quad (40)$$

The above statement holds true if $f(n) = n^x$ where $x \geq 1$.

7. Determine the asymptotic order of the expression $\sum_{i=1}^n a^i$ where $a > 0$ is a constant, i.e. find a simple function $g(n)$ such that the expression is in the class $\Theta(g(n))$. (Hint: consider the cases $a = 1$, $a > 1$, and $0 < a < 1$ separately.)

Solution. If $a > 1$ then:

$$\sum_{i=1}^n a^i \leq \sum_{i=1}^n a^n \Rightarrow n \cdot a^n \quad (41)$$

$$a^1 + a^2 + \dots + a^n \leq a^n + a^n + \dots + a^n = n \cdot a^n = O(n \cdot a^n) \quad (42)$$

$$a^1 + a^2 + \dots + a^n \geq a^{\frac{n}{2}} + a^{\frac{n}{2}+1} + \dots + a^n = \frac{n}{2} \cdot a^n = \frac{\Omega(n \cdot a^n)}{2} \quad (43)$$

$$\Theta(n \cdot a^n) = O(n \cdot a^n) \cap \frac{\Omega(n \cdot a^n)}{2} \quad (44)$$

If $a = 1$ then:

$$\sum_{i=1}^n a^i = \sum_{i=1}^n 1^i = \Theta(n * 1) = \Theta(n) \quad (45)$$

If $0 < a < 1$ then let $a = 0.5$:

$$\sum_{i=1}^n a^i = \sum_{i=1}^n 0.5^i = 0.5^1 + 0.5^2 + \dots + 0.5^n = \sum_{i=1}^n a^i \in \Theta(\log(n)) \quad (46)$$

8. Use induction to prove that $\sum_{k=1}^n k^4 = \frac{n(n+1)(6n^3+9n^2+n-1)}{30}$ for all $n \geq 1$

Solution. *Base:*

$$n = 1 \quad (47)$$

$$1^4 = \frac{1(1+1)(6(1)^3+9(1)^2+(1)-1)}{30} \quad (48)$$

$$\Rightarrow 1 = 1$$

Inductive step:

$$\begin{aligned} f(n+1) &= \sum_{k=1}^{n+1} k^4 = 1^4 + 2^4 + 3^4 \dots n^4 + (n+1)^4 \quad (49) \\ &= \frac{(n+1)((n+1)+1)(6(n+1)^3+9(n+1)^2+(n+1)-1)}{30} \\ &\Rightarrow \frac{n(n+1)(6n^3+9n^2+n-1)}{30} + (n+1)^4 = \frac{(n+1)(n+2)(2 \cdot n+3)(3 \cdot n^2+9 \cdot n+5)}{30} \\ &\Rightarrow \frac{(n+1)(2 \cdot n+3)(n+2)(3 \cdot n^2+9 \cdot n+5)}{30} = \frac{(n+1)(n+2)(2 \cdot n+3)(3 \cdot n^2+9 \cdot n+5)}{30} \end{aligned}$$

The above result is True therefore $f(n+1)$ holds true for all $n \geq 1$. Therefore $\sum_{k=1}^n k^4 = \frac{n(n+1)(6n^3+9n^2+n-1)}{30}$ also holds true for all $n \geq 1$.